

24-01-2026

Python - 4

- Discussion on resource •
- Non-Primitive Data Types •

Non-primitive Data types

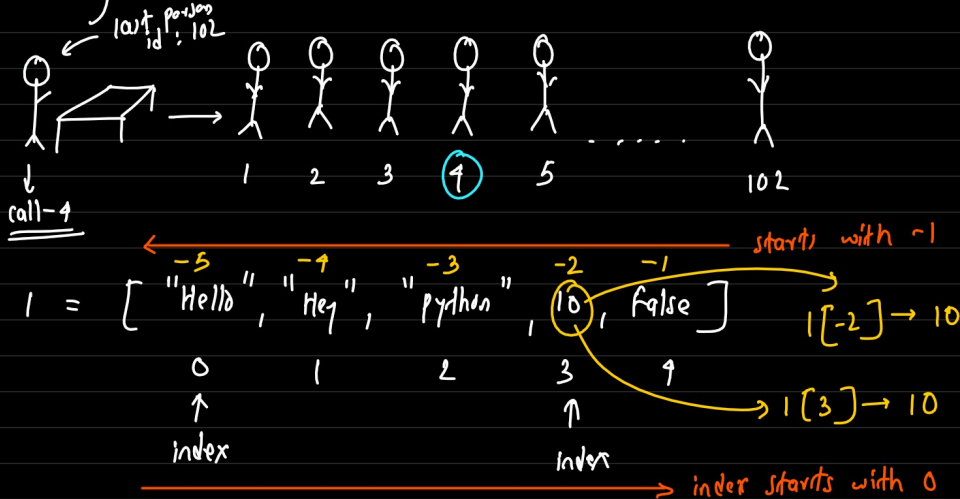
$l = [0] \rightarrow l = [0, 2, 2]$

$l = []$, $l.append(0)$, $append(2)$, $append(2)$

List :

- 1) When we append something to the list, it maintain the order to append.
- 2) Accepts any data type, duplicates are allowed.
- 3) Methods :
 - append() : add an element/value to the end
 - pop() : remove by index

Accessing list elements/values



$l[3] \rightarrow$ index-3 of list `l`
 $\rightarrow 10$

Q. problem statement: i want to know the last value of value.

$\rightarrow$ get the first value $\rightarrow$ index-0

$\rightarrow$ get the last value $\rightarrow$ index-?

index $\rightarrow -1$

$\downarrow$

last element

["Hello", "Hey", "python", 10, false]

len() ← string → count its characters

len() ← list → count total number of elements / value

↓
total number of elements

1, 2, 3 → ③ index ≠ no. of elements
↑ ↑ ↑
0 1 2

len("Data")

a = "Data" → len(a) → len("Data")

append() → add element

len() → count

index → used to access value of list

pop() → remove element from list()

[~~0~~, 1, 2, 3]

[0, 0, 0]

→ [0, 0, 0, 0 ... 0] → 1000's element

↪ remove "Hey"

pop(index)

append(value)

remove(value) → remove the first occurrence

[0, 10, 5, 4]

append(4) ↗

insert() → 4 → index 1

→ [0, 4, 10, 5, 4]

0 1 2 3 4

list : —

append(), index-concept, pop(), insert(), remove()

Dictionary :

Dictionary (The key-value pair) :

- 1) To store data with labels (keys) instead of positions
- 2) Key must be (String/Numbers/Tuples) values can be anything.
- 3) Keys should always be unique, duplicate values are accepted.

person ↔ Dob, phone, city, pob, occupation
 ↑
 labelled values

register →	person	Dob	phone	city	pob	occupation
	Monal	-	-	-	-	-
	Rohit	-	-	-	-	-
→	prideen	-	-	-	-	-

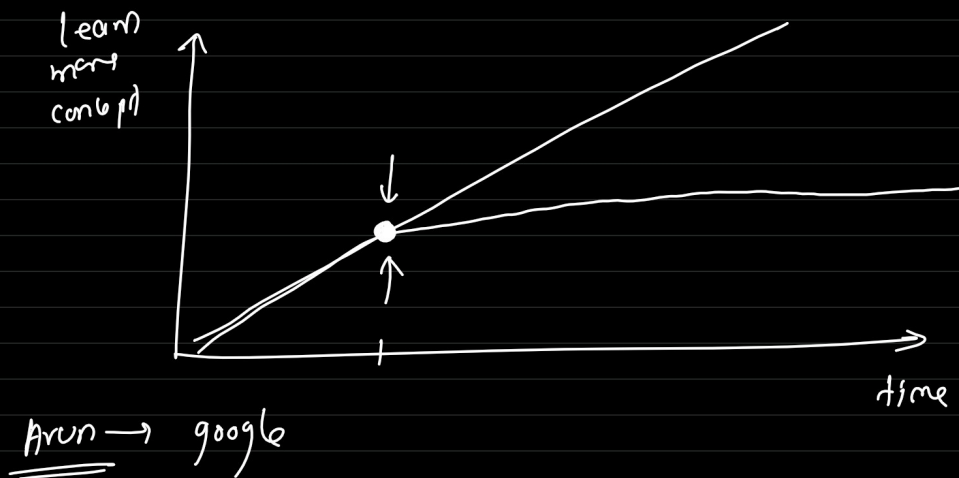
register = { } → curly bracket

register = { "prideen" : [dob, phone, city, pob, occupation] }
 key : value

syntax → { key : value }

$r = \left\{ \begin{array}{l} \text{"prithvi"} : [\text{job, phone, city, pob, occupation}], \\ \text{"rohit"} : [\text{job, phone, city, pob, occupation}], \\ \text{"Ankit"} : [\text{job, phone, city, pob, occupation}], \\ \text{"Amit"} : [\text{job, phone, city, pob, occupation}] \end{array} \right\}$

$r[\text{"Amit"}] \rightarrow [\text{job, phone, city, pob, occupation}]$
 $\uparrow$
 key



Register :

task: I want to note down details of visitors.

Name	Address	ph.no	ID
Mona	—	—	—
x	—	—	—
y	—	—	—
z	—	—	—

Mona $\rightarrow$ name

monal = [Address, ph.no, ID]
1000' → lots of variable

register = [["monal", Address, ph.no, ID], [], ["amir"]]
→ this solves the problem but not efficient

dictionary = { key : value }

= { "monal": [Address, ph.no, ID],
"amir": [Address, phone] }

dict()

→ .keys() → all the keys in a dict

→ .values() → all the values in a dict

→ .get() → provide key to get value

a = [1, 2, 3, 4]

[101, 102, 103, 104] .keys()

↙
a = [[], [], [], []]
↑ ↑ ↑ ↑

$$a = [1, 2, 3, 4] \cdot \text{keys}()$$

$$a = [[], [], [], []] \cdot \text{value}()$$

mutable $\rightarrow$ changes are allowed
 immutable $\rightarrow$ changes are not allowed

list = [] $\rightarrow$ append, pop $\rightarrow$ mutable

Tuple (Immutable) :

- 1) Data that should not change (ID)
- 2) Prevents accidental cases where data can be overwritten.
- 3) It accepts anything
- 4) Limitation : append(), pop(), no re-assignment after creation.

$$\text{dob} = (10, 'March', 1920)$$

$$\text{dob}[0] = 20$$

$$\text{Bangalore} = (x.1229, y.2329) \rightarrow \text{tuple}$$

$$a = 1, 2, 3, 4 \rightarrow a = [1, 2, 3, 4]$$

$$[x, y, \downarrow, \downarrow, \downarrow]$$

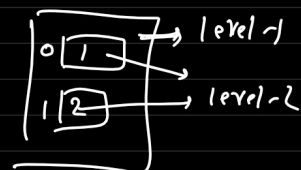
key: -

$$l = [1, 2, \textcircled{3}, 4]$$

0 1 2 3

$$l[2] \rightarrow 3$$

$$l_2 = \left[\underset{\substack{\uparrow \\ \text{level-1}}}{[1, 2, 3, 4]}, \underset{\substack{\uparrow \\ \text{level-2}}}{[5, 6, 7, \textcircled{8}]} \right]$$



[illegible]
$$1-2[1] \begin{bmatrix} -1 \\ 1 \end{bmatrix} \rightarrow 8$$

```

{
  "Name" : {
    "First_name" : "Monal",
    "Last_name" : "S"
  },
  "DOB" : (10, "feb", 1920),
  "Phone.no" : "+91-1111111111"
}

```

→ register[101] →
 → register.get(101)

```

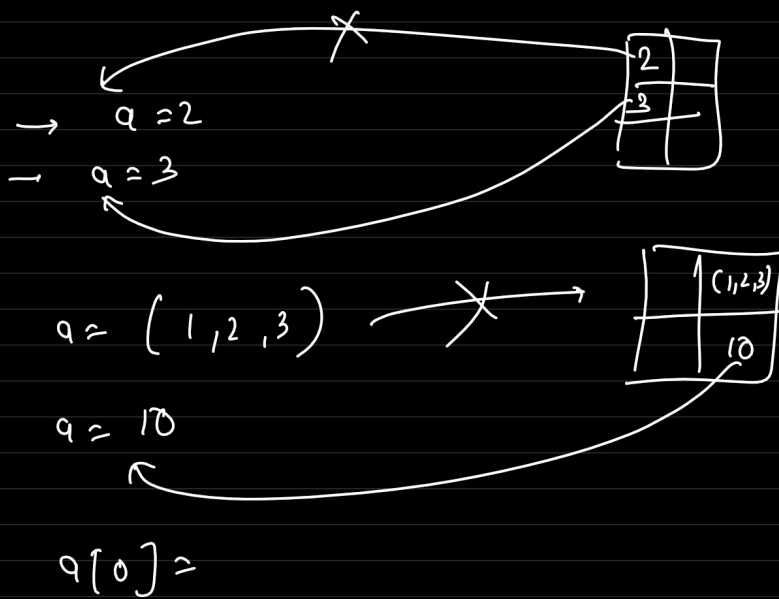
register_v3 = {
  101 : {
    "Name" : {
      "First_name" : "Monal",
      "Last_name" : "S"
    },
    "DOB" : (10, "feb", 1920),
    "Phone.no" : "+91-1111111111"
  }
}

```

↓
 register[101]["DOB"]
 ↓
 (10, "feb", 1920)

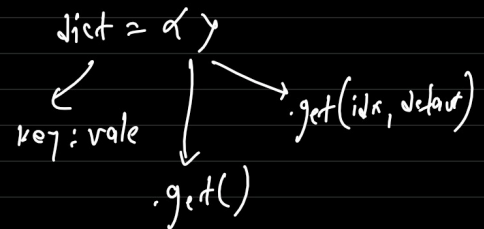
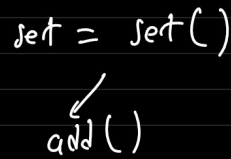
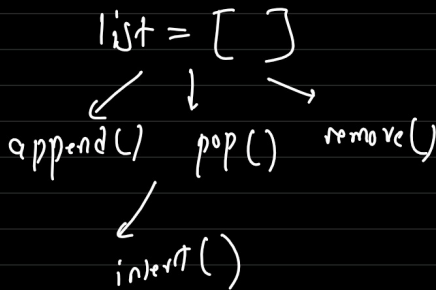
if not able to understand $\rightarrow$ * watch recording
* practice

if you are able to understand $\rightarrow$ * practice



SET (The unique values) :

- 1) To handle uniqueness
- 2) Removes duplicates automatically
- 3) NOTE : Unordered



$tuple = tuple()$
 $= ()$

$len() \rightarrow \text{function}$
 $\rightarrow \text{str}$
 $\rightarrow \text{iterable}$

$sorted \rightarrow list$
 $\rightarrow set$
 $o/p \rightarrow list$